

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

*I S MD Abdur Rahman Ghouse(192311197) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: S MD Abdur Rahman Ghouse

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and**

trailer. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers.

Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

SHORT ANSWER SOLUTIONS:

1. File transfer with Data Link overhead

File size = 150 MB = 150,000,000 bytes

Segment size = 1500 bytes

Number of segments/frames = $150,000,000 \div 1500 = 100,000$ frames

Data Link overhead = $100,000 \times 40 = 4,000,000$ bytes = 4 MB

Total data transmitted = 150 MB + 4 MB = 154 MB

Transmission time = $(154 \times 8) \div 50 = 24.64$ seconds

2. Sliding window transmission

Window size = 6 frames

Total frames = 48

Number of windows = $48 \div 6 = 8$ windows

One frame lost per window, so retransmissions = 8

Total transmissions = $48 + 8 = 56$ transmissions

3. IP fragmentation

Packet size = 12,000 bytes

MTU = 1500 bytes

IP header = 20 bytes

Data per fragment = $1500 - 20 = 1480$ bytes

Number of fragments = $12,000 \div 1480 = 9$ fragments

Total header overhead = $9 \times 20 = 180$ bytes

4. Sliding window transmission

This is the same as Question 2.

Number of windows = $48 \div 6 = 8$ windows

Retransmissions = 8

Total transmissions = $48 + 8 = 56$ transmissions

5. Session Layer synchronization

File size = 600 MB

Checkpoint interval = 60 MB

Failure occurs after = 390 MB

Checkpoints before failure:

60, 120, 180, 240, 300, 360 MB = 6 checkpoints

Last checkpoint = 360 MB

Data to retransmit = $390 - 360 = 30$ MB

6. Presentation Layer compression and encryption

Original file = 900 MB

Compression = 40%

Compressed size = $900 \times 60\% = 540$ MB

Encryption overhead = 5% of 540

Encrypted size = $540 \times 1.05 = 567$ MB

Reduction = $900 - 567 = 333$ MB

Percentage reduction = $(333 \div 900) \times 100 = 37\%$

7. FTP application transfer

Data size = 3 GB = 3000 MB

Request size = 3 MB

Number of requests = $3000 \div 3 = 1000$ requests

Data in megabits = $3000 \times 8 = 24,000$ Mb

Transmission time = $24,000 \div 120 = 200$ seconds

8. End-to-end delay

Processing delays:

Application = 4 ms

Transport = 3 ms

Network = 5 ms

Data Link = 2 ms

Physical = 1 ms

Total processing delay = $4 + 3 + 5 + 2 + 1 = 15$ ms

Propagation delay = 18 ms

Transmission delay = 12 ms

Total delay = $15 + 18 + 12 = 45$ ms

9. Frame overhead and efficiency

Useful data = 80 MB = 80,000,000 bytes

Frame size = 1600 bytes

Overhead per frame = 80 bytes

Useful data per frame = $1600 - 80 = 1520$ bytes

Number of frames = $80,000,000 \div 1520 = 52,632$ frames

Total overhead = $52,632 \times 80 = 4,210,560$ bytes ≈ 4.21 MB

Transmission efficiency = $(1520 \div 1600) \times 100 = 95\%$

10. Healthcare network transmission

Original data = 240 MB

Compression = 30%

Compressed size = $240 \times 70\% = 168$ MB

Compressed data in bytes = 168,000,000 bytes

Segment size = 1400 bytes

Number of segments = $168,000,000 \div 1400 = 120,000$ segments

Data Link overhead = $120,000 \times 60 = 7,200,000$ bytes = 7.2 MB

Final data transmitted = 168 MB + 7.2 MB = 175.2 MB